PARABOLIC

- CONVERGENZA $\quad u_h \xrightarrow[h\to 0]{} u^{ex}$

u_h: SOLUZIONE DEL METODO DISCRETO

$h = \Delta x$

$h, \Delta t$?

$h \to 0, \quad \Delta t \to 0$

$\Delta t = C\,\Delta x$

$\Delta t = C \cdot \Delta x^2$

$\lambda_i \sim \frac{1}{\Delta x^2} \qquad |\Delta t\, \lambda_i| \le 2$

$\infty \xleftarrow[\Delta x \to 0]{} \frac{\Delta x}{\Delta x^2}|\alpha|$

$\xleftarrow{\Delta t = C\Delta x} \frac{\Delta t}{\Delta x^2}|\alpha| \le 2$

$\Delta t = C\Delta x^2$

$C\frac{\Delta x^2}{\Delta x^2}|\alpha| \le 2$

STABILITÀ

$$(*)\quad U^{n+1} = B(\Delta t)\cdot U^n + b^n(\Delta t) \qquad U^{n+1}, U^n, b^n \in \mathbb{R}^{N_x}$$

$N_x \sim \frac{1}{\Delta x}$ $\qquad B(\Delta t) \in \mathbb{R}^{N_x \times N_x}$

- E.E. $\quad B(\Delta t) = I + \Delta t A$
- E.I. $\quad B(\Delta t) = (I - \Delta t A)^{-1}$ $\quad$ (MATRICE DI DERIVATA SECONDA)

- C.N. $B(\Delta t) = \left(I - \frac{\Delta t}{2} A\right)^{-1} \left(I + \frac{\Delta t}{2} A\right)$

STABILITÀ DI LAX-RICHTMYER

IL METODO (*) è stabile se

per $\forall T > 0$ TEMPO FINALE $\exists C_T > 0$

: $\forall \Delta t$ e $\forall n$: $n \cdot \Delta t \leq T$

$\Rightarrow \quad \| B(\Delta t)^n \| \leq C_T$.

TEOREMA EQUIVALENZA DI LAX

- UN METODO CONSISTENTE e LINEARE DI TIPO (*)

$\Rightarrow$ CONVERGENTE $\Leftrightarrow$ STABILE.

CHECK STABILITÀ DI LAX-RICHTMYER

- EULERO IMPLICITO

$$\| B(\Delta t) \|_2 = \max \operatorname{eig}\left((I - \Delta t A)^{-1}\right) \leq 1$$

$$\lambda_i(A) \leq 0$$

$$\lambda_i(-A) \geq 0$$

$$\lambda_i(I - \Delta t A) \geq 1$$

$$\lambda_i\left((I - \Delta t A)^{-1}\right) \leq 1$$

$$\|B(\Delta t)^n\|_2 \le \|B(\Delta t)\|_2^n \le 1^n = 1 = C_T$$

$\Rightarrow$ EULERO IMPLICITO È STABILE

C.N. $B(\Delta t) = \left(I - \frac{\Delta t}{2} A\right)^{-1} \left(I + \frac{\Delta t}{2} A\right)$

$$\|B(\Delta t)\|_2 \le \max_i \frac{1 + \frac{1}{2}\Delta t\, \lambda_i(A)}{1 - \frac{1}{2}\Delta t\, \lambda_i(A)} \le 1$$

perché $\lambda_i(A) \le 0$

C.N. $\Rightarrow$ STABILE

- EULERO ESPLICITO

$$\|B(\Delta t)\| \le 1 + \alpha\, \Delta t$$

$$\alpha \sim \frac{1}{\Delta x^2}$$

$$\|B(\Delta t)\|^n \le (1 + \alpha\, \Delta t)^n \le e^{\alpha T}$$

- VON NEUMANN STABILITY ANALYSIS

SI BASA SU FOURIER

- BC PERIODICHE
- COEFF eq. COSTANTI

- FUNZIONI $\underbrace{e^{ikx}}$ SONO AUTOFUNZIONI DI ∂_x

- DISCRETAMENTE $\underline{W_j^k} = e^{i(j\cdot\Delta x)\cdot k}$ $\forall k \in \mathbb{Z}$

es.

$$(DV)_j = \frac{V_{j+1} - V_{j-1}}{2\Delta x}$$

$x_j = j \cdot \Delta x$

$j = 1, \dots, N_x$

$x \in [0, 2\pi]$

$$(DW^k)_j = \frac{1}{2\Delta x}\left[e^{i(j+1)\cdot\Delta x k} - e^{i(j-1)\Delta x k}\right]$$

$$= \underbrace{e^{ij\Delta x k}} \frac{i}{\Delta x}\left[\frac{e^{i\Delta x k} - e^{-i\Delta x k}}{2i}\right]$$

$$= W_j^k \underbrace{\frac{i}{\Delta x}\sin(\Delta x \cdot k)}$$

NON DIPENDE DA j

PERÒ DIPENDE DA $k \in \mathbb{Z}$

$$\|U^{n+1}\|_2 \le (1+\alpha\Delta t)\cdot\|U^n\|_2$$

α INDIP DA Δt?

$$U^{n+1} = \sum_k c_k^{n+1} \underbrace{W^k}_{\text{BASE ORTONORMALE}}$$

$$\|c_k^{n+1}\|_2 \le (1+\alpha\Delta t)\,\|c_k^n\|_2$$

$$c_k^{n+1} = \underbrace{g(k)}_{\text{NON DIPENDE DA } j,\ N_x} \cdot c_k^n \qquad U^{n+1} = A U^n$$

DIPENDE DA $k \in \mathbb{Z}$

$$|g(k)| \leq 1 + \alpha \Delta t$$

<u>EULERO ESPLICITO</u> esempio

FISSIAMO $k \in \mathbb{Z}$ $(\forall k \in \mathbb{Z})$

$$c_k^{n+1} \cdot e^{ijk\Delta x} = c_k^n e^{ijk\Delta x}$$

$$+ \frac{\Delta t}{\Delta x^2}\left(c_k^n e^{i(j+1)\Delta x \cdot k} - 2 c_k^n e^{i(j)\cdot \Delta x \cdot k} + c_k^n e^{i(j-1)\Delta x \cdot k}\right)$$

$$U_j^{n+1} = U_j^n + \frac{\Delta t}{\Delta x^2}\left(U_{j+1}^n - 2U_j^n + U_{j-1}^n\right)$$

$$c_k^{n+1} \cancel{e^{ijk\Delta x}} = \cancel{e^{ijk\Delta x}} \cdot c_k^n \left\{ 1 + \frac{\Delta t}{\Delta x^2}\left[e^{ik\Delta x} - 2 + e^{-ik\Delta x}\right]\right\}$$

$$c_k^{n+1} = g(k) \cdot c_k^n$$

$$g(k) = 1 + \frac{\Delta t}{\Delta x^2}\left(e^{ik\Delta x} + e^{-ik\Delta x} - 2\right)$$

$$= 1 + \frac{\Delta t}{\Delta x^2}\left(2\cos(k\Delta x) - 2\right)$$

$$= 1 + \frac{\Delta t}{\Delta x^2} 2 \underbrace{\left(\cos(k\Delta x) - 1\right)}_{\underbrace{-2 \le \quad \le 0}_{-4 \le \quad \le 0}}$$

DIFFICILE ✓

$$1 - 4\frac{\Delta t}{\Delta x^2} \le g(k) \le 1$$

$$|g(k)| \le 1$$

$$\rightarrow 1 - 4\frac{\Delta t}{\Delta x^2} \ge -1$$

$$4\frac{\Delta t}{\Delta x^2} \le 2$$

$$\underbrace{\Delta t \le \frac{\Delta x^2}{2}}$$ STABILITÀ PER E.E.

VON NEUMANN ANALYSIS X EULERO IMPLICITO DIFF. FINITE CENTRATE IN SPAZIO

$$U_j^{n+1} = U_j^n + \Delta t \frac{U_{j+1}^{n+1} - 2U_j^{n+1} + U_{j-1}^{n+1}}{\Delta x^2}$$

$$c_k^{n+1} \cdot e^{ij\Delta x k} = c_k^n \cdot e^{ij\Delta x k} + \frac{\Delta t}{\Delta x^2} c_k^{n+1} \cdot [\ldots$$

$$\dots e^{i(j+1)\Delta x k} - 2e^{ij\Delta x k} + e^{i(j-1)\Delta x k}\Big]$$

$$\cancel{e^{ij\Delta x k}}\, c_k^{n+1}\left[1 - \frac{\Delta t}{\Delta x^2}\left(e^{i\Delta x k} - 2 + e^{-i\Delta x k}\right)\right]$$

$$= c_k^n\, \cancel{e^{ij\Delta x k}}$$

$$g(k) = \Bigg(\overbrace{\Big[\underbrace{1 - \frac{\Delta t}{\Delta x^2}2\big(\cos(k\cdot\Delta x) - 1\big)}_{-2 \le \Downarrow \le 0}\Big]}^{\ge 1 \;>0}\Bigg)^{-1} \le 1$$

$|g(k)| \le 1$?

SÌ

$\Rightarrow$ STABILE SEMPRE $\forall \Delta t, \Delta x > 0$

$$c_k^1(t) = \underbrace{\overbrace{-\frac{1}{\Delta x^2}2\big(1 - \cos(\Delta x\cdot k)\big)}^{\lambda_k}} \cdot c_k(t)$$

Se $\lambda_k \in S \quad \forall k \in \mathbb{Z}$

$\Downarrow$

REGIONE DI STABILITÀ DEL METODO RK

$\Rightarrow$ STABILITÀ

- FORMA DEBOLE

FORTE

$$\begin{cases} \partial_t u - a\Delta u(x,t) = f(t,x) & \\ u(0,x) = u_0(x) & x \in \Omega \\ u(x,t) = u_D(x) & x \in \Gamma_D \\ a\partial_n u(x,t) = g_N(x) & x \in \Gamma_N \end{cases}$$

$V \in H^1_{\Gamma_D}(\Omega)$ TROVA $u(t) \in H^1$ $\forall t$

+ BC DIRICHLET

$$\begin{cases} \int_\Omega \partial_t u(x,t) \cdot v(x)\, dx + \int_\Omega a\nabla u(x,t)\, \nabla v(x)\, dx \\ = \int_\Omega f(t,x) \cdot v(x)\, dx + \int_{\Gamma_N} a\nabla u(x,t) \cdot n \cdot v(x)\, d\Gamma \\ u(t,x) = u_D(x) \quad \forall x \in \Gamma_D \end{cases}$$

$$\begin{cases} \int \partial_t u(t,x) \cdot v(x)\, dx + a(u(t), v) = F(v;t) \\ u(t,x) = u_D(x) \quad x \in \Gamma_D \\ u(t=0,x) = u_0(x) \quad x \in \Omega \end{cases}$$

- $a(\cdot,\cdot) : V \times V \to \mathbb{R}$ è BILINEARE, LIMITATA, DEBOLMENTE COERCITIVA

$\exists\, \lambda \geq 0, \alpha \geq 0 : \quad a(u,u) + \underbrace{\lambda \|u\|^2_{L^2}} \geq \alpha \|u\|^2_V$

- F LINEARE CONTINUO
- $u_0 \in L^2$ $\quad f \in L^2(\mathbb{R}^+ \times \Omega)$

$\Rightarrow \exists!$ u SOLUZIONE DEL PROBLEMA DEBOLE

STIMA SULL'ENERGIA

- a COERCIVA
- DIRICHLET OMOGENEO
- OPPURE PERIODICO

$$\int_\Omega v\, \partial_t u + a(u,v) = 0$$

$v = u \in V$

$$\int_\Omega u \cdot \partial_t u + a(u,u) = 0$$

$$\int_\Omega \partial_t \frac{u^2}{2} + a(u,u) = 0$$

$$\partial_t \int_\Omega \frac{u^2}{2} = - a(u,u) \leq -\alpha \cdot \|u\|_V^2 < 0$$

$\Rightarrow \quad \|u(t)\|_{L^2} \leq \|u(0)\|_{L^2}$

FINITE ELEMENT

$$u_h(x,t) = \sum \varphi_i(x) \cdot u_i(t)$$

$$\int_\Omega \sum_i \partial_t u_i(t)\, \varphi_i(x) \cdot \varphi_j(x)\, dx$$

$$+ \sum_i u_i(t) \cdot a(\varphi_i, \varphi_j) = F(\varphi_j)$$

$$= \int \nabla \varphi_i \nabla \varphi_j$$

$$M \, \partial_t \underline{u} + A \cdot \underline{u} = \underline{f}$$

es.

- EULERO IMPLICITO

$$\underline{\underline{M}} \cdot \underline{U}^{n+1} - \underline{\underline{M}} \cdot \underline{U}^{n} + \Delta t \cdot \underline{\underline{A}} \cdot \underline{U}^{n+1} = \underline{f}^{n}$$

$$(\underline{\underline{M}} + \Delta t \, \underline{\underline{A}}) \, \underline{U}^{n+1} = \underline{\underline{M}} \, \underline{U}^{n} + \underline{f}^{n}$$

$\forall \; n = 1, ___, N_t$ SISTEMA LINEARE

NON MATRICE COSTANTE IN TEMPO

QUINDI POSSIAMO FATTORIZZARE $(M + \Delta t A)$ 1 VOLTA SOLA E POI UTILIZZARE LA FORMA FATTORIZZATA X RISOLVERE I N_t SYS LINEARI.

STABILITÀ ENERGETICA X E.I.

$$\int (u_h^{n+1} - u_h^{n}) \cdot \underbrace{u_h^{n+1}}_{\text{TEST}} + \Delta t \, \nabla u_h^{n+1} \cdot \nabla u_h^{n+1} \, dx = 0$$

$$\|u_h^{n+1}\|_{L^2}^2 + \Delta t \, |u_h^{n+1}|_{H^1} = \int u_h^{n} \cdot u_h^{n+1}$$

CAUCHY-SCHWARZ $\leq \|u^{n}\|_2 \cdot \|u^{n+1}\|_2$

$$\leq \frac{\|u^n\|_2^2}{2} + \frac{\|u^{n+1}\|_2^2}{2}$$

$$0 < (a \pm b)^2 = a^2 \pm 2ab + b^2$$

$$\frac{a^2}{2} + \frac{b^2}{2} > -ab$$

$$> ab$$

$$\frac{\|u^{n+1}\|_2^2}{2} + \Delta t\, |u_h^{n+1}|_{H^1} \leq \underbrace{\frac{\|u^n\|_2^2}{2}}_{\text{ENERGIA}(t^n)}$$

$$\leq \underbrace{\frac{\|u^{n+1}\|_2^2}{2}}_{\text{ENERGIA}(t^{n+1})}$$

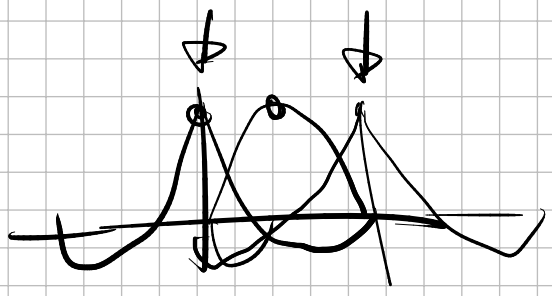
SETTIMANA NO LEZIONE

$\Rightarrow$ FATE SCHEDA SU DIFFERENZE FINITE + EQUAZIONE DEL CALORE.

PENSATE AI PROGETTI

$$\underline{U_j^{n+1}} = \frac{\underline{U_{j+1}} - 2\underline{U_j} + \underline{U_{j-1}}}{\Delta x^2}$$

$\mathbb{R}^2$



$e^{i\Delta x(j+1)k}$

$$c_k^{n+1} = g(k)\, c_k^n$$

$$\begin{pmatrix} c_k^E \\ c_k^C \end{pmatrix} = \begin{pmatrix} g_{11}^{(k)} & g_{12}^{(k)} \\ g_{21}^{(k)} & g_{22}^{(k)} \end{pmatrix} \begin{pmatrix} c_k^{E\,n} \\ c_k^{C\,n} \end{pmatrix}$$